

Supplementary Material

Reproducibility of Pollution–Health Evidence Synthesis Using LLM-Assisted Screening and Extraction

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1 PubMed Search Strategy

All abstracts were retrieved from PubMed via the NCBI E-utilities API (`esearch` + `efetch`) on February 11, 2026. Two complementary queries were used:

1.1 Broad Query (573 records)

Listing 1: PubMed broad search query

```
("Particulate Matter"[MeSH] OR "PM2.5" OR "fine particulate matter"
OR "fine particles" OR "ambient particulate matter")
AND ("Hospitalization"[MeSH] OR "hospital admission"
OR "hospital admissions" OR "emergency department"
OR "ED visit" OR "ED visits")
AND ("Respiratory Tract Diseases"[MeSH] OR "respiratory"
OR "asthma" OR "COPD" OR "pneumonia" OR "bronchitis"
OR "bronchiolitis")
AND ("time series" OR "time-series" OR "case-crossover"
OR "distributed lag" OR "Poisson regression" OR "GAM"
OR "generalized additive model")
```

1.2 Design-Filtered Query (222 records)

Listing 2: PubMed design-filtered query

```
("PM2.5" OR "fine particulate matter"
OR "particulate matter 2.5")
AND ("respiratory" OR "asthma" OR "COPD" OR "pneumonia")
AND ("hospitalization" OR "hospital admission"
OR "emergency department")
AND ("time series" OR "time-series" OR "case-crossover")
```

1.3 Exclusion Candidates (475 records)

A third query targeted articles clearly outside scope (animal studies, in-vitro, occupational exposure) to populate the “clearly exclude” category.

1.4 Corpus Assembly

From the merged candidate pool (1,021 unique records after deduplication):

- 100 abstracts meeting all six inclusion criteria were labeled “clearly include”
- 100 abstracts with unambiguous exclusion reasons were labeled “clearly exclude”
- 300 abstracts with borderline characteristics (e.g., mixed pollutants, ambiguous outcomes) were labeled “ambiguous”

The 200 “clear” abstracts were labeled by an automated classification rule based on inclusion-criteria keyword matching, and the 300 “ambiguous” abstracts by the same rule operating in low-confidence regions (see Labeling Guide, Section 4). No abstract in the 500-item corpus received a human label at this stage; human dual-independent labeling was conducted later, on a stratified 100-abstract subset, and is reported in Section 4.

2 Prompt Templates

The exact prompts used for both pipeline stages are reproduced below. These prompts were held constant across all 6 models and all 120 experimental runs.

2.1 Stage A: Abstract Screening Prompt

Listing 3: Screening prompt template (configs/prompts/screening.txt)

```
You are an expert epidemiologist conducting a systematic review on
the association between PM2.5 exposure and respiratory
hospitalizations.

Given the following title and abstract of a scientific publication,
determine whether it should be INCLUDED or EXCLUDED from the review
based on the criteria below.

## Inclusion Criteria
1. Original epidemiological study (not a review, editorial, or
   commentary)
2. Exposure: PM2.5 or fine particulate matter (<=2.5 um)
3. Outcome: respiratory hospitalization or emergency department visit
4. Study design: time-series or case-crossover
5. Reports a relative risk (RR), odds ratio (OR), or equivalent
   effect measure with confidence interval
6. Published in English

## Exclusion Criteria
- Reviews, meta-analyses, editorials, or commentaries
- Exposure other than PM2.5 (unless PM2.5 is one of multiple
  exposures analyzed)
- Outcome other than respiratory hospitalization/ED visit
- No quantitative effect estimate reported
- Animal or in-vitro studies

## Input
Title: {title}
Abstract: {abstract}

## Instructions
Respond with a JSON object only. Do not include any text outside
the JSON.

{
  "decision": "include" | "exclude" | "uncertain",
  "confidence": 0.0-1.0,
  "rationale": "brief explanation of decision",
  "exposure": "PM2.5" | "other" | "not_specified",
  "outcome": "respiratory_hospitalization" | "respiratory_ED"
             | "other" | "not_specified",
  "study_design": "time_series" | "case_crossover" | "cohort"
                  | "other" | "not_specified",
  "has_effect_estimate": true | false
}
```

2.2 Stage B: Data Extraction Prompt

Listing 4: Extraction prompt template (configs/prompts/extraction.txt)

```
You are an expert epidemiologist extracting quantitative data from a
scientific publication on PM2.5 and respiratory hospitalizations for
a systematic review and meta-analysis.

Given the following title and abstract, extract the structured data
below. If a field is not reported in the abstract, use null.

## Input
Title: {title}
Abstract: {abstract}

## Instructions
Extract ALL reported effect estimates. If multiple estimates are
reported (e.g., different lags, subgroups), extract each as a
separate entry in the "estimates" array.

Respond with a JSON object only. Do not include any text outside
the JSON.

{
  "study_id": "first_author_year",
  "study_location": "city, country",
  "study_period": "YYYY-YYYY",
  "study_design": "time_series" | "case_crossover" | "other",
  "population": "general" | "elderly" | "children" | "other",
  "sample_size": null | integer,
  "estimates": [
    {
      "effect_measure": "RR" | "OR" | "HR" | "IRR" | "other",
      "effect_estimate": float,
      "ci_lower": float,
      "ci_upper": float,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3" | "per IQR" | "other",
      "lag": "lag0" | "lag1" | "lag0-1" | "other",
      "outcome_specific": "all_respiratory" | "pneumonia"
                          | "COPD" | "asthma" | "other",
      "covariates": ["temperature", "humidity", "trend",
                    "seasonality", "..."]
    }
  ]
}
```

3 JSON Schemas for Output Validation

All LLM outputs were validated against JSON Schema (Draft-07) definitions. Outputs failing validation were flagged but retained to ensure complete provenance.

3.1 Screening Output Schema

Seven required fields, no additional properties permitted: `decision` (enum: include/exclude/uncertain), `confidence` (0.0–1.0), `rationale` (string, min 10 chars), `exposure`, `outcome`, `study_design` (categorical enums), and `has_effect_estimate` (boolean).

3.2 Extraction Output Schema

Required fields: `study_id` (string), `estimates` (array). Optional metadata: `study_location`, `study_period`, `study_design`, `population`, `sample_size`. Each estimate requires: `effect_measure`, `effect_estimate`, `ci_lower`, `ci_upper` (all with range constraints 0.1–10.0).

Full schemas are available in the repository at `configs/schemas/`.

4 Gold Standard Labeling Protocol

4.1 Screening Labels

The 500-item corpus was labeled by an automated classification rule: the 200 “clear” abstracts (100 include, 100 exclude) by inclusion-criteria keyword matching, and the 300 “ambiguous” abstracts by the same rule in low-confidence regions. The rule made no errors on a held-out 50-abstract validation set, which by the rule of three bounds its true error rate at approximately 6% with 95% confidence; we do not describe it as 100% precise. These labels serve as reference values for accuracy computation but do not affect the primary metric (EMR), which measures self-consistency regardless of ground truth.

4.2 Dual-Independent Human Validation

In response to peer review we validated the automated labels against human judgement on a stratified subset of 100 abstracts (25 clear-include, 25 clear-exclude, 50 ambiguous; stratified by category $\times$ year-tertile, seed=42).

Pre-registration. The protocol, the screening templates, and the agreement script were frozen as an OSF registration on 2026-05-12 (<https://doi.org/10.17605/OSF.IO/FGN3E>). The two label sets were returned on 2026-07-15 and 2026-07-29, so the registration pre-dates all label collection. The registration pre-specified the target (Cohen’s $\kappa \geq 0.80$, Cochrane Handbook §4.6.6), the discordance-rate expectation ($< 15\%$), and a contingency for missing the target.

Both pre-specified gates were missed, and we report them rather than drop them: $\kappa = 0.529$ against a target of 0.80, and a discordance rate of 25% against an expectation of $< 15\%$.

Two competing explanations for a low coefficient can be ruled out. The prevalence index is low (0.189) and PABAK is almost identical to κ , so the value is not depressed by skewed marginals—the “kappa paradox” is not operating. Weighted variants shift the estimate by less than 0.02, so the result is not an artefact of the three-class scale. With $n = 100$ the interval spans “fair” through “substantial”, so we report the value relative to the threshold and do not assign it a descriptive band.

The disagreement is directional, not diffuse. The binary discordant cells are 2 versus 19 (exact McNemar $p = 2.2 \times 10^{-4}$), and marginal homogeneity is rejected on the 3-class table (Stuart–Maxwell $\chi^2(2) = 15.4$, $p = 4.6 \times 10^{-4}$; Bowker symmetry likewise rejected). Rater 2 returned 49 INCLUDE decisions against rater 1’s 30. Systematic directional divergence of this kind indicates that the two raters applied different readings of the written protocol, which is a defect of the instrument rather than noise in the raters.

Table 1: Stage-A screening agreement between the two human raters ($n = 100$). Analytic intervals use the Fleiss–Cohen–Everitt delta method; the bootstrap uses 10,000 resamples with seed 42.

Quantity	Value	95% CI
Cohen’s κ , 3-class	0.529 (SE 0.074)	[0.383, 0.674]
bootstrap, percentile	0.529	[0.383, 0.670]
bootstrap, BCa	0.529	[0.382, 0.669]
Cohen’s κ , binary (include vs. exclude, $n = 95$)	0.556	[0.400, 0.712]
Weighted κ , linear (ordinal include–uncertain–exclude)	0.541	—
Weighted κ , quadratic	0.548	—
Raw agreement	75.0%	—
PABAK, 3-class / binary	0.500 / 0.558	—
Prevalence index / bias index	0.189 / 0.179	—
Test against Cochrane target	$z = -3.65$	$p = 1.3 \times 10^{-4}$

Table 2: Confusion matrix, rater 1 (rows) $\times$ rater 2 (columns).

	INCLUDE	EXCLUDE	UNCERTAIN
INCLUDE	28	2	0
EXCLUDE	19	46	2
UNCERTAIN	2	0	1

Of the 25 discordances, 19 run in that same direction, and 17 of those 19 turn on inclusion criterion 5 (quantitative effect estimate). Protocol v1.1 stated the criterion as “reports RR, OR, HR (or equivalent) with 95% CI” in its inclusion list while its exclusion list read “no quantitative effect estimate extractable from the abstract” — the first wording admits a stated intention to report an effect, the second demands the numbers themselves. Compounding this, the v1.1 decision table specified outcomes for failure in two or more criteria and for a borderline single criterion, but not for clear failure in exactly one criterion, which is the configuration of those 17 abstracts.

Table 3: Stratum-specific agreement. In the clear-exclude stratum expected agreement equals one, so κ is undefined despite unanimity; PABAK is reported for that reason.

Stratum	n	Raw agreement	Cohen’s κ	PABAK
clear-exclude	25	1.000	undefined	1.000
clear-include	25	0.680	0.359	0.360
ambiguous	50	0.660	0.398	0.320

The gold standard is asymmetrically valid, and this qualifies every accuracy figure in the main text. The exclude side is unanimous: both raters endorsed exclusion for all 25 abstracts the automated rule called clearly excludable. The include side is not: of the 25 abstracts the rule called clearly includable, rater 1 endorsed inclusion for 13 and rater 2 for 21, and 8 of the 25 discordances fall inside that stratum. Reported specificity therefore rests on labels the humans confirm, while reported sensitivity rests on include-side labels they substantially contest and should be read as a lower bound.

Protocol amendment (v1.2). Following item (c) of the registered contingency, criterion 5 was split into three levels. Level 5a: a numeric point estimate with a numeric interval is present, and the criterion is satisfied. Level 5b: the abstract states that the effect was estimated but omits the values, which resolves to *uncertain*, on the reasoning that abstract-only screening cannot verify what

the abstract omits and that an EXCLUDE would assert something about the study the available evidence does not support. Level 5c: no estimate and no mention of one, which resolves to *exclude*, because total absence is reasonable evidence that the study produced no extractable estimate. The distinction between 5b and 5c is not which criterion failed but how much the abstract reveals about the failure.

The decision table now also separates structural criteria (original study, PM_{2.5} exposure, respiratory hospitalization, English), where a single clear failure excludes, from conditional criteria (design, effect reporting), where a single failure yields *uncertain* — with level 5c as the stated exception. Precedence when several rows apply is exclude > uncertain > include. The amended table is exhaustive by design: the v1.1 defect was an uncovered case, and raters are now instructed to record *uncertain* and escalate rather than improvise if they meet one.

Recalibration round and its limits. A blinded round restricted to the 25 discordant items is underway; the 75 concordant items are not reopened. This round cannot produce a second estimate of κ , and we do not report one. Conditioning re-measurement on prior disagreement moves a recomputed whole-corpus coefficient upward mechanically—at a sufficiently high resolution rate it would cross the Cochrane threshold by construction, carrying no evidential content. The outcome is therefore reported as post-hoc reconciliation agreement conditional on the initially discordant items, and $\kappa = 0.529$ stands as the study’s agreement estimate. Demonstrating that the v1.2 amendment repairs the ambiguity would require a fresh independent sample, which we identify as the appropriate next step rather than claim here.

Discordance resolution. Items still discordant after recalibration go to a consensus meeting; items without consensus are settled by the senior author (Y.d.S.T.) as tie-breaker, as pre-registered. The role is deliberately held by a co-author who did not develop the pipeline being scored against this gold standard, which keeps final adjudication independent of the system under evaluation. Tie-breaks are restricted to residual discordant items, adjudicated against the written criteria with no access to model outputs, and logged with the criterion invoked, so each adjudicated item is auditable. The resulting gold standard records, per abstract, whether its label came from first-round agreement, post-recalibration agreement, consensus, or tie-break.

4.3 Extraction Labels

For the 100 “clearly include” articles, extraction templates were created following standardized rules:

- Prefer the primary/main estimate if multiple are reported
- Prefer all-respiratory over specific conditions (asthma, COPD)
- Prefer single-pollutant model over multi-pollutant
- Prefer lag 0–1 if multiple lags reported without a “best” designation
- Record per 10 $\mu\text{g}/\text{m}^3$ – convert if different increment used
- If abstract reports % change, convert: $\text{RR} = 1 + (\% \text{change}/100)$

Discordance threshold: > 0.005 for effect estimates, > 0.01 for CI bounds.

5 Example Outputs: Exact Match vs. Near-Miss

To illustrate the nature of extraction non-determinism, we present two representative cases.

5.1 Example 1: Exact Match (Local Model – Gemma 2 9B)

All 10 runs produced identical output (EMR = 1.000):

Listing 5: Gemma 2 9B extraction output (identical across all 10 runs)

```
{
  "study_id": "chen_2017",
  "study_location": "Shanghai, China",
  "study_period": "2013-2015",
  "study_design": "time_series",
  "population": "general",
  "sample_size": null,
  "estimates": [
    {
      "effect_measure": "RR",
      "effect_estimate": 1.023,
      "ci_lower": 1.005,
      "ci_upper": 1.042,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3",
      "lag": "lag0-1",
      "outcome_specific": "all_respiratory",
      "covariates": ["temperature", "humidity", "day_of_week",
                    "long_term_trend", "seasonality"]
    }
  ]
}
```

5.2 Example 2: Near-Miss (Cloud API – Claude Sonnet 4.5)

Runs 1–8 and 10 produced one output; Run 9 differed in the `estimates` array:

Listing 6: Claude Sonnet 4.5 – Run 9 extracted 2 estimates instead of 1

```
{
  "study_id": "chen_2017",
  "study_location": "Shanghai, China",
  "study_period": "2013-2015",
  "study_design": "time_series",
  "population": "general",
  "sample_size": null,
  "estimates": [
    {
      "effect_measure": "RR",
      "effect_estimate": 1.023,
      "ci_lower": 1.005,
      "ci_upper": 1.042,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3",
      "lag": "lag0-1",
      "outcome_specific": "all_respiratory",
      "covariates": ["temperature", "humidity", "day_of_week",
                    "long_term_trend", "seasonality"]
    },
  ],
}
```

```

{
  "effect_measure": "RR",
  "effect_estimate": 1.031,
  "ci_lower": 1.008,
  "ci_upper": 1.054,
  "ci_level": 95,
  "exposure_increment": "per 10 ug/m3",
  "lag": "lag2",
  "outcome_specific": "pneumonia",
  "covariates": ["temperature", "humidity", "day_of_week"]
}
]
}

```

Key observation: The metadata fields (`study_id`, `study_location`, `study_design`) are identical across all 10 runs (consistent with BERTScore $F_1 = 0.997$). The variation lies in the `estimates` array—Run 9 extracted an additional sub-group estimate (pneumonia at lag 2) that the other 9 runs did not. This single additional estimate causes the hash-based EMR to register a non-match, while the BERTScore over metadata text remains near-perfect. This exemplifies the “semantic paradox” reported in Section 4.5 of the main paper.

6 Model Configurations

Table 4: Complete model configurations for all six LLMs.

Model	Provider	Temp.	Seed	Max Tokens	Timeout	Version
LLaMA 3 8B	Ollama	0.0	42	4096	600s	llama3:8b
Mistral 7B	Ollama	0.0	42	4096	600s	mistral:7b
Gemma 2 9B	Ollama	0.0	42	4096	600s	gemma2:9b
Claude Sonnet 4.5	Anthropic	0.0	—	4096	90s	claude-sonnet-4-5-20250929
Gemini 2.5 Pro	Google	0.0	42	8192 [†]	90s	gemini-2.5-pro
GPT-4.1	OpenAI	0.0	42	4096	90s	gpt-4.1

[†]Gemini requires higher `maxOutputTokens` because thinking tokens consume the output budget.

6.1 Provider API Documentation Links

For full reproducibility and to disambiguate the exact API surfaces used in this study, Table 5 lists the official documentation pages for each provider and the local serving runtime, with the access date for each link. This is provided so that future reviewers and replicators can verify (a) exactly which parameters were exposed on each API surface at the time of the study, (b) which parameters were renamed, deprecated, or added subsequently, and (c) whether default values (e.g., for unset `top_p`, `stop_sequences`, or system prompts) have changed silently between the study window and any future replication attempt.

Access dates correspond to the start of data collection for each stack. The full request payload schemas used in this study, including provider-specific defaults explicitly set or accepted, are reproduced as JSON skeletons in our public repository (`configs/request_payloads/`).

Table 5: Official documentation references for the six deployment stacks evaluated and the desconfound cloud route. URLs and access dates are recorded so that replication studies can pin to the documented behaviour rather than to current docs that may drift.

Provider	Stack / End-point	Documentation URL	Access date
Anthropic	Claude Sonnet 4.5 / Messages API v1	https://docs.anthropic.com/en/api/messages	2026-02-11
Google	Gemini 2.5 Pro / generateContent (v1)	https://ai.google.dev/api/rest	2026-02-11
OpenAI	GPT-4.1 / Chat Completions v1	https://platform.openai.com/docs/api-reference/chat	2026-02-11
Ollama (local)	LLaMA 3 8B / Mistral 7B / Gemma 2 9B (v0.15.5)	https://github.com/ollama/ollama/blob/main/docs/api.md	2026-02-11
OpenRouter → DeepInfra	meta-llama/llama-3.1-8B / openrouter.ai/docs (desconfound route)	https://openrouter.ai/docs	2026-04-23

7 Provenance Protocol Details

7.1 Call-Level Hashing

Each LLM call produces a **call hash** computed as:

Listing 7: Call hash computation (src/provenance/hasher.py)

```
import hashlib, json

def compute_call_hash(prompt, input_text, model_id, temperature,
                      seed=None):
    fields = {
        "prompt": prompt,
        "input_text": input_text,
        "model_id": model_id,
        "temperature": temperature,
        "seed": seed,
    }
    canonical = json.dumps(fields, sort_keys=True,
                           ensure_ascii=True)
    return hashlib.sha256(canonical.encode()).hexdigest()
```

The use of `sort_keys=True` and `ensure_ascii=True` ensures deterministic serialization across platforms and Python versions.

7.2 Output Hashing

The raw LLM response text is hashed directly:

```
def compute_output_hash(output_text):
    return hashlib.sha256(output_text.encode()).hexdigest()
```

7.3 Run Cards

Each run produces a structured JSON run card containing:

- Run metadata: model ID, provider, stage, run number
- Configuration: temperature, seed, max tokens
- Execution: timestamps, call counts (total/success/fail), duration
- Provenance: aggregate output hash (SHA-256 of sorted individual hashes joined by pipe delimiter)

7.4 Verification Protocol

If two runs of the same model produce different aggregate hashes, the per-call output hashes are compared to identify exactly which abstracts changed. This enables targeted analysis of non-determinism at the individual-abstract level.

8 Corpus Statistics

Table 6: Corpus descriptive statistics.

Statistic	Value
Total abstracts	500
Unique journals	148
Publication range	1994–2026
Mean abstract length (characters)	1,873
Median abstract length (characters)	1,812
Min abstract length (characters)	423
Max abstract length (characters)	4,291
Clear include	100 (20%)
Clear exclude	100 (20%)
Ambiguous	300 (60%)

9 Experimental Timeline

All 120 experimental runs were conducted within a 3-week window to minimize the risk of silent API model updates:

10 Software and Dependencies

- **Python:** 3.14.3
- **OS:** macOS Darwin 24.6.0 (Apple M4, 24 GB RAM)
- **Ollama:** v0.15.5 (local model serving)

Table 7: Experiment execution timeline.

Model	Screening	Extraction	Duration
LLaMA 3 8B	Feb 11–12	Feb 12–13	~91h
Mistral 7B	Feb 13–15	Feb 15–16	~77h
Gemma 2 9B	Feb 16–17	Feb 17	~46h
Claude Sonnet 4.5	Feb 18	Feb 18–19	~12h
Gemini 2.5 Pro	Feb 20	Feb 20–21	~8h
GPT-4.1	Feb 22	Feb 22–Mar 2	~6h

- **Key dependencies:** `numpy` ≥ 1.26 , `pandas` ≥ 2.1 , `scipy` ≥ 1.12 , `jsonschema` ≥ 4.20 , `bert-score` $\geq 0.3.13$, `transformers` ≥ 4.36
- **No external SDKs:** All API calls use Python’s `urllib.request` to ensure uniform request handling
- **Test suite:** 45 tests (`pytest` ≥ 8.0), 100% pass rate
- **License:** MIT
- **Repository:** <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>

11 Complete Bootstrap Confidence Intervals

All confidence intervals were computed using the percentile bootstrap method with 10,000 resamples. For models with $\text{EMR} = 1.000$, the bootstrap trivially returns $[1.000, 1.000]$; the rule-of-three upper bound on the non-match rate is $3/n$ (0.6% for $n = 500$ screening, 3.0% for $n = 100$ extraction).

Table 8: Complete EMR with 95% bootstrap confidence intervals for all model-stage combinations.

Model	Stage	EMR	95% CI	n
LLaMA 3 8B	Screening	1.000	$[1.000, 1.000]^*$	480
LLaMA 3 8B	Extraction	1.000	$[1.000, 1.000]^*$	96
Mistral 7B	Screening	1.000	$[1.000, 1.000]^*$	500
Mistral 7B	Extraction	1.000	$[1.000, 1.000]^*$	100
Gemma 2 9B	Screening	1.000	$[1.000, 1.000]^*$	500
Gemma 2 9B	Extraction	1.000	$[1.000, 1.000]^*$	100
Claude Sonnet 4.5	Screening	0.974	$[0.960, 0.986]$	500
Claude Sonnet 4.5	Extraction	0.050	$[0.010, 0.090]$	100
Gemini 2.5 Pro	Screening	0.936	$[0.914, 0.956]$	500
Gemini 2.5 Pro	Extraction	0.200	$[0.120, 0.280]$	100
GPT-4.1	Screening	0.970	$[0.954, 0.984]$	500
GPT-4.1	Extraction	0.150	$[0.080, 0.220]$	100

*Rule-of-three upper bound on non-match rate: $3/n$ (screening: 0.6%, extraction: 3.0%).

12 Fixed-Slot Extraction Prompt and Comparison

The fixed-slot prompt template used in the sensitivity analysis (main text §4.7):

Listing 8: Fixed-slot extraction prompt (configs/prompts/extraction_fixed_slot.txt)

```

You are an expert epidemiologist extracting quantitative data from a
scientific publication on PM2.5 and respiratory hospitalizations for
a systematic review and meta-analysis.

Given the following title and abstract, extract the SINGLE PRIMARY effect
estimate using the fixed selection rules below. Do not produce a
variable-length array; produce exactly one primary_estimate object (or
null if no estimate is reportable).

## Fixed selection rules (apply in order)
1. Prefer all-respiratory outcomes over specific subgroups.
2. Prefer single-pollutant PM2.5 model over multi-pollutant adjustments.
3. If multiple lags are reported, prefer lag 0-1 (cumulative); else lag 0.
4. Prefer per 10 ug/m^3 exposure increment; convert if reported per IQR.
5. If percent change reported, convert: RR = 1 + (%change/100).
6. If no quantitative effect estimate with CI is reportable, set
    primary_estimate = null.

## Input
Title: {title}
Abstract: {abstract}

## Output (JSON only, no extra text)
{
  "study_id": "first_author_year",
  "study_location": "city, country",
  "study_period": "YYYY-YYYY",
  "study_design": "time_series" | "case_crossover" | "other",
  "population": "general" | "elderly" | "children" | "other",
  "sample_size": null | integer,
  "primary_estimate": { ... } | null,
  "n_estimates_in_abstract": integer
}

```

Table 9: Fixed-slot extraction sensitivity analysis: 3 cloud models $\times$ 3 runs $\times$ 100 INCLUDE articles. Variable-length baseline from main Table 3 (10 runs). Provider pinned per call: Claude Sonnet 4.5 via Anthropic API; Gemini 2.5 Pro via Google API; GPT-4.1 via OpenAI API.

Model	Var-len EMR	Fixed-slot EMR	Δ abs	Δ rel	Pairwise dis.
Claude Sonnet 4.5	0.050	0.030	−0.020	−40.0%	0.78
Gemini 2.5 Pro	0.200	0.185	−0.015	−7.6%	0.49
GPT-4.1	0.150	0.444	+0.294	+196.3%	0.36

13 LLaMA Cloud Desconfound Experiment Details

The desconfound experiment (main text §4.8) serves the identical model weights via cloud and compares against the local Ollama deployment.

Table 10: Desconfound experimental configuration. Both deployments use the SAME model weights, prompt, seed=42, and temperature=0; only the serving infrastructure differs.

Parameter	Local (Ollama)	Cloud (DeepInfra)
Model	llama3:8b	meta-llama/llama-3-8b-instruct
Provider	Ollama v0.15.5	OpenRouter → DeepInfra (pinned)
Hardware	Apple M4, 24 GB	DeepInfra GPU cluster
Inference	Single-stream	Batched continuous-batching
Seed	42	42
Temperature	0.0	0.0
Max tokens	4,096	512 (screening)
N runs	10	10
N abstracts	500	500
Date range	Feb 11–13, 2026	Apr 25–28, 2026

Table 11: Per-run successful-call counts. Cloud runs 5 and 6 experienced transient infrastructure failures resulting in reduced valid-item counts, which we report transparently as a cloud-reliability finding. Local runs are uniformly 480/500 successful (the same 20 abstracts deterministically exceed response length in every local run).

Run	Local valid	Cloud valid
1	480	491
2	480	489
3	480	491
4	480	489
5	480	45*
6	480	221*
7	480	491
8	480	491
9	480	491
10	480	491

*Transient cloud-side failure (503 Service Unavailable bursts).

14 Silver Standards and Cross-Validation

Two independent silver standards complement the heuristic gold standard for the 100 INCLUDE articles.

14.1 Silver-Internal: Majority Vote of Six Evaluated Models

Consensus extraction is computed as the per-field mode across all 6 models $\times$ 10 runs (60 votes per item). Numeric fields are binned at 0.01 resolution before mode computation.

Table 12: Silver-internal consensus. Items with consensus = items where at least one model-run produced a non-null value. Mean mode-agreement = average proportion of votes matching the modal value across items with consensus.

Field	Items w/ consensus	Mean mode-agreement
effect_estimate	95/100	0.858
ci_lower	84/100	0.846
ci_upper	84/100	0.845
effect_measure	100/100	0.783
outcome_specific	100/100	0.694
exposure_increment	100/100	0.761
lag	100/100	0.644

14.2 Silver-External: DeepSeek-R1 (5 Runs)

DeepSeek-R1 (`deepseek-reasoner`) is a reasoning model from a different provider family than the 6 evaluated models, providing an independent reference. Five runs were aggregated by per-field majority vote.

Table 13: Silver-external consensus from DeepSeek-R1 (5 runs $\times$ 100 items).

Field	Items w/ consensus	Mean mode-agreement
effect_estimate	84/100	0.890
ci_lower	74/100	0.894
ci_upper	74/100	0.889
effect_measure	84/100	0.906
outcome_specific	84/100	0.803
exposure_increment	84/100	0.881
lag	75/100	0.862

For items lacking DeepSeek-R1 consensus across its 5 runs (16 of 100 items on `effect_estimate`; 26 of 100 items on the CI bounds `ci_lower/ci_upper`; 25 of 100 on `lag`), per-run accuracy against silver-external is undefined and these items are excluded; the accuracies reported in the corresponding silver-external comparison tables (and the cross-validation table below) are computed over the consensus subset only.

Table 14: Field-level agreement between silver-internal and silver-external (numeric tolerance 0.05; categorical exact match). Spearman ρ on `effect_estimate` = **0.845**.

Field	n compared	Agreement
effect_estimate	84	0.798
ci_lower	74	0.797
ci_upper	74	0.784
effect_measure	84	0.738
outcome_specific	84	0.655
exposure_increment	84	0.798
lag	75	0.653

14.3 Cross-Validation Between Silvers

15 Pairwise Disagreement and Rationale Similarity

Table 15: Pairwise disagreement (mean fraction of $\binom{10}{2}=45$ run-pairs producing different outputs per item; 0 = perfect determinism). Compare to EMR (Table 10): EMR collapses items into binary match/non-match while pairwise disagreement reports the dispersion within non-matches.

Model	Stage	EMR	Pairwise	% items any-disagree
LLaMA 3 8B	screening	1.000	0.0000	0.00%
LLaMA 3 8B	extraction	1.000	0.0000	0.00%
Mistral 7B	screening	1.000	0.0000	0.00%
Mistral 7B	extraction	1.000	0.0000	0.00%
Gemma 2 9B	screening	1.000	0.0000	0.00%
Gemma 2 9B	extraction	1.000	0.0000	0.00%
Claude Sonnet 4.5	screening	0.974	0.0096	2.60%
Claude Sonnet 4.5	extraction	0.050	0.8180	95.00%
Gemini 2.5 Pro	screening	0.936	0.0198	6.40%
Gemini 2.5 Pro	extraction	0.200	0.4927	80.00%
GPT-4.1	screening	0.970	0.0137	3.00%
GPT-4.1	extraction	0.150	0.4858	85.00%

16 BERTScore F1: Raw vs. Rescaled (Random-Pair Baseline)

BERTScore F1 with `roberta-large` is known to saturate near 1.0 even for thematically unrelated sentence pairs because the underlying contextual embeddings cluster all natural-English text in a small subspace. The package’s authors recommend rescaling F1 against a per-language random-pair baseline (`rescale_with_baseline=True, lang="en"`) to make absolute magnitudes interpretable. We compute both raw and rescaled F1 over the same $\binom{10}{2}=45$ run-pairs per article (4,500 pairs per model).

The rescaling substantially separates the API models: raw F1 places all three within 0.003 of each other (saturation), whereas rescaled F1 reveals that Gemini’s minimum pair similarity drops

Table 16: Token-bigram Jaccard similarity on screening rationale (free-form text) vs. metadata (categorical concatenation). Computed across all $\binom{10}{2}=45$ run-pairs per item, averaged over 500 abstracts. Demonstrates that BERTScore $F_1 \geq 0.997$ on metadata is a saturation artifact: in non-saturating text (rationale), Claude exhibits only 0.378 mean similarity.

Model	Metadata Jaccard	Rationale Jaccard	Gap
LLaMA 3 8B	1.000	1.000	0.000
Mistral 7B	1.000	1.000	0.000
Gemma 2 9B	1.000	1.000	0.000
Claude Sonnet 4.5	0.978	0.378	0.600
Gemini 2.5 Pro	0.978	0.623	0.355
GPT-4.1	0.984	0.725	0.259

Table 17: BERTScore F1 (raw vs. rescaled-with-baseline) per deployment stack, computed on five-field metadata concatenations across all 4,500 run-pairs. Rescaled F1 subtracts the random-pair baseline, exposing the genuine semantic agreement signal that raw F1 saturates. Negative rescaled values (e.g., Gemini’s minimum -0.458) indicate pairs less similar than random English text. Means/std/min/max are over the 4,500 pairs.

Deployment Stack	Raw F1			Rescaled F1		
	Mean	Min	≥ 0.95	Mean	Min	≥ 0.95
LLaMA 3 8B / Ollama / M4	1.000	1.000	100.0%	1.000	1.000	100.0%
Mistral 7B / Ollama / M4	1.000	1.000	100.0%	1.000	1.000	100.0%
Gemma 2 9B / Ollama / M4	1.000	1.000	100.0%	1.000	1.000	100.0%
Claude Sonnet 4.5 / Anthropic	0.997	0.931	99.7%	0.983	0.594	88.7%
Gemini 2.5 Pro / Google	0.997	0.754	98.5%	0.979	-0.458	93.4%
GPT-4.1 / OpenAI	1.000	0.956	100.0%	0.998	0.740	98.9%

below the random-English baseline (-0.458), confirming that some Gemini extraction-text pairs are genuinely semantically dissimilar—not merely subject to numeric-precision noise. This supports the main-text observation (Finding 3 of Section 4.5) that “Gemini shows the lowest minimum BERTScore $F_1=0.754$. . . some articles receive genuinely divergent interpretations across runs.”

17 Multiple Comparison Correction for Field-Level Contrasts

The original two-proportion z -test was replaced with McNemar’s test because the contrasts compare *paired* per-item consistency outcomes on the same 100 articles (or 500 abstracts) across cloud APIs. We compute, for each model M and field F , the article-level binary indicator $b_i^{M,F}=1$ if all 10 runs of M produced an identical value for F on article i , 0 otherwise; the paired contrasts are then evaluated by McNemar’s test on the 2×2 discordance table of $(b_i^{M_1,F}, b_i^{M_2,F})$, using the exact mid- p binomial when discordants $b+c\leq 25$ and the chi-square with continuity correction otherwise (`statsmodels.stats.contingency_tables.mcnemar`). Holm-Bonferroni (FWER) and Benjamini-Hochberg (FDR) corrections are applied to the 21 paired contrasts (5 metadata fields plus overall extraction and screening EMR; 3 pairwise comparisons each). Of 21 contrasts: 7 are significant at raw $p<0.05$, 4 survive Holm correction, and 5 survive BH-FDR.

Table 18: Paired McNemar contrasts of cloud-API consistency surviving Holm-Bonferroni or BH-FDR correction at $\alpha=0.05$. “ b/c ” are the discordant cell counts ($b=M_1$ match, M_2 mismatch; $c=M_1$ mismatch, M_2 match).

Field	Models	Δ EMR	b/c	Raw p	Holm p	BH-FDR p
study_location	Claude vs. GPT-4.1	-0.220	0/22	<0.0001	<0.0001	<0.0001
study_location	Gemini vs. GPT-4.1	-0.104	0/10	0.0020	0.0391	0.0136
extraction_overall_EMR	Claude vs. Gemini	-0.150	4/19	0.0026	0.0468	0.0136
screening_overall_EMR	Claude vs. Gemini	$+0.038$	27/8	0.0023	0.0446	0.0136
study_location	Claude vs. Gemini	-0.115	2/13	0.0074	0.1255	0.0310

The remaining 16 contrasts (population, sample_size, study_design, study_period; some screening/extraction overall pairs) are not statistically robust after FWER correction; the original “distinct instability profiles” claim (main text Finding 3) is therefore tightened to: *differences are robust for study_location, overall extraction EMR (Claude vs. Gemini), and overall screening EMR (Claude vs. Gemini), but not for other categorical fields*. Replacing the original two-proportion z -test with McNemar slightly increases the number of contrasts surviving Holm correction (from 3 to 4), as McNemar conditions on the discordant pairs and gains power when within-pair concordance is high.

18 Fleiss’ Kappa for Inter-Run Agreement

We report Fleiss’ κ for inter-run agreement on screening decisions and extraction outputs, treating each of the 10 LLM runs as one rater. Fleiss’ κ adjusts the raw concordance for the agreement expected by chance under fixed-marginal multinomial sampling, providing a more interpretable benchmark than EMR alone (which conflates “perfectly consistent” with “trivially consistent in a degenerate-marginals regime”).

Two caveats govern how the table below should be read, and we state them before the numbers.

First, the chance correction is inert for the whole-output hash column. Treating each distinct output hash as a category yields 667 categories over 100 items for Claude, 265 for Gemini and 298

for GPT-4.1. With categories that numerous and that sparsely populated, the expected agreement p_e falls to $\approx 0.003\text{--}0.005$, so $\kappa=(p_o-p_e)/(1-p_e)$ collapses onto p_o and reproduces the pairwise concordance already reported in Table 7 rather than adding chance-corrected information. We keep the column because the ordering across stacks is informative, and we do not interpret it against the Landis–Koch bands, which presuppose a meaningful chance baseline. The per-field columns, where categories number in the single digits, are not affected.

Second, the intervals are bootstrap, not closed-form. The classical variance of Fleiss (1971, eq. 13) depends only on the marginal proportions and is the asymptotic variance *under* H_0 : $\kappa=0$, appropriate for testing that null and not for placing an interval around an estimated κ . Applied to these data it returned upper limits above 1.000 in 18 of 24 cells and assigned different standard errors to stacks whose κ was exactly 1.000. We therefore report percentile bootstrap intervals resampling items (10,000 resamples, seed 42). Where every resample returns $\kappa=1.000$ the interval is degenerate and is shown as $[1.000, 1.000]$; the informative statement for those cells is the rule-of-three bound on the non-match rate, as used for the $\kappa=1.000$ cells in the main text.

Table 19: Fleiss’ κ (95% CI) for inter-run agreement across 10 runs per deployment stack. Screening: 3-class decisions (include/exclude/uncertain) over items present in all 10 runs. Extraction (hash): each unique output hash treated as a distinct category. Extraction (per categorical field): study_design and population. Landis & Koch (1977) reference bands: < 0.0 poor, $0.0\text{--}0.2$ slight, $0.21\text{--}0.4$ fair, $0.41\text{--}0.6$ moderate, $0.61\text{--}0.8$ substantial, $0.81\text{--}1.0$ almost perfect; these bands are not applied to the hash column, where the chance baseline is inert (see text). Intervals are percentile bootstrap over items (10,000 resamples, seed 42). [†]every bootstrap resample returned $\kappa=1.000$; the interval is degenerate and the rule-of-three bound on the non-match rate is the informative quantity.

Deployment Stack	Screening (3-class)	Extraction (hash)	study_design	population
LLaMA 3 8B / Ollama / M4	1.000 [†]	1.000 [†]	1.000 [†]	1.000 [†]
Mistral 7B / Ollama / M4	1.000 [†]	1.000 [†]	1.000 [†]	1.000 [†]
Gemma 2 9B / Ollama / M4	1.000 [†]	1.000 [†]	1.000 [†]	1.000 [†]
Claude Sonnet 4.5 / Anthropic	0.981 [0.970, 0.991]	0.180 [0.133, 0.226]	0.985 [0.952, 1.000]	0.974 [0.944, 1.000]
Gemini 2.5 Pro / Google	0.975 [0.961, 0.987]	0.505 [0.445, 0.559]	1.000 [†]	0.979 [0.956, 1.000]
GPT-4.1 / OpenAI	0.974 [0.960, 0.986]	0.511 [0.453, 0.564]	0.990 [0.970, 1.000]	0.997 [0.989, 1.000]

For screening decisions, all three local stacks attain the trivial upper bound $\kappa=1.000$ and the three cloud APIs fall in the “almost perfect” band ($\kappa\in[0.974, 0.981]$). We deliberately avoid the shorthand that the stacks therefore “reach the Cochrane guideline”: inter-run Fleiss’ κ measures one stack reproducing itself across repetitions, whereas the Cochrane $\kappa\geq 0.80$ guideline concerns agreement between two independent human reviewers. On this corpus our own two human raters reached $\kappa=0.529$ (Section 4.2), and Mistral-7B attains perfect self-consistency at a specificity of 0.240, which together show that the two quantities cannot be substituted for one another. For *whole-output* extraction hashes, the coefficient falls to 0.180 (Claude), 0.505 (Gemini) and 0.511 (GPT-4.1). Because the chance baseline is inert at this granularity, these figures should be read as the observed whole-output concordance rather than as chance-corrected agreement, and we do not attach Landis–Koch labels or compare them to the Cochrane $\kappa\geq 0.80$ threshold. The ordering is nonetheless the substantive point, and it matches the EMR ranking: Claude’s 10 runs reproduce a whole extraction output far less often than either other cloud stack, and none of the three approaches the local stacks. Per-field κ on categorical extraction fields (study_design, population) remains in the “almost perfect” band for all stacks, isolating the source of low whole-output κ to the variable-

length `estimates` array (already documented in Section 4.4 of the main text).

19 Hartung-Knapp-Sidik-Jonkman Random-Effects Sensitivity

we re-ran the per-run random-effects meta-analysis (full 100-article corpus, one pooled estimate per model×run combination) using the Hartung-Knapp-Sidik-Jonkman (HKSJ) variance correction alongside the original DerSimonian-Laird (DL) Wald- z formulation. Both estimators share the same τ^2 (method-of-moments); HKSJ inflates the standard error to $\sqrt{q^*/\sum w_i^*}$ and uses $t_{0.975,k-1}$ as the critical value. Cochrane Handbook 6.5+ recommends HKSJ over DL for $k < 10$ because DL is known to be liberal (anti-conservative).

Table 20: Per-run random-effects pooled RR (mean across 10 runs) and 95% CIs under DL (Wald- z) vs. HKSJ (t_{k-1}), with HKSJ-to-DL width ratio. “% runs cross null” = fraction of the 10 LLM runs whose pooled 95% CI on the full 100-article corpus includes RR=1.

Deployment Stack	Mean RE RR	Mean DL CI	Mean HKSJ CI	HKSJ/DL width	% runs cross null
LLaMA 3 8B	1.0377	[1.031, 1.045]	[1.019, 1.057]	2.65×	0% (DL) / 0% (HKSJ)
Mistral 7B	1.0317	[1.025, 1.038]	[1.016, 1.048]	2.51×	0% (DL) / 0% (HKSJ)
Gemma 2 9B	1.0253	[1.020, 1.031]	[1.012, 1.039]	2.51×	0% (DL) / 0% (HKSJ)
Claude Sonnet 4.5	1.0148	[1.013, 1.017]	[1.009, 1.021]	2.70×	0% (DL) / 0% (HKSJ)
Gemini 2.5 Pro	1.0213	[1.018, 1.025]	[1.012, 1.031]	2.54×	0% (DL) / 0% (HKSJ)
GPT-4.1	1.0236	[1.019, 1.028]	[1.013, 1.034]	2.22×	0% (DL) / 0% (HKSJ)

On the full 100-article corpus, HKSJ widens CIs by 2.2–2.7× but every per-run CI continues to exclude RR=1 under both estimators. Thus the main-text conclusion that “no run-pair within any model produced a 95% random-effects CI crossing the null in the full-corpus pooling” is *robust* to the choice of DL or HKSJ. The HKSJ correction matters substantially in the small-literature simulation (Section 20), where the t_{k-1} critical value and the multiplicative q^* inflation become dominant.

20 Small-Literature Meta-Analytic Simulation

HKSJ sensitivity. we re-ran the small-literature simulation with the Hartung-Knapp-Sidik-Jonkman (HKSJ) variance correction alongside the original DL/Wald- z formulation. HKSJ uses the same method-of-moments τ^2 estimator as DL but inflates the standard error to $\sqrt{q^*/\sum w_i^*}$, where $q^* = \frac{1}{k-1} \sum w_i^* (y_i - \hat{\theta}_{RE})^2$, and replaces the $z_{0.975}$ Wald critical value with $t_{0.975,k-1}$. Because t -percentiles inflate as $k \rightarrow 2$ and $q^* \geq 1/\sum w_i^*$ under heterogeneity, HKSJ produces substantially wider CIs in small literatures. The unstable-null-crossing rate under HKSJ is markedly higher: at $k=10$, **31.5%** of Claude subsamples, **48.5%** of Gemini subsamples, and **20.0%** of GPT-4.1 subsamples produce run-dependent reversals of the meta-analytic conclusion (versus 0.5%, 0.5%, and 0.0% under DL). The qualitative conclusion of the main text—*LLM non-determinism propagates into the downstream conclusion only in small literatures*—is robust to the choice of variance estimator; the magnitude of the effect is substantially amplified under HKSJ, which Cochrane Handbook 6.5+ recommends precisely for $k < 10$.

Table 21: Random-effects pooled RR variation across 200 random subsamples per k , seed=42, computed independently for each of 10 LLM runs. “Unstable null-crossing” = subsamples where the pooled 95% CI crosses RR=1 in some runs but not others (i.e., the meta-analytic conclusion changes depending on the LLM run). Both DerSimonian-Laird (DL, Wald-z) and Hartung-Knapp-Sidik-Jonkman (HKSJ, t_{k-1}) variances are reported; HKSJ is recommended by Cochrane Handbook 6.5+ for $k < 10$.

Model	k	Mean range	P95 range	% unstable (DL)	% unstable (HKSJ)
Claude Sonnet 4.5	10	0.0082	0.0292	0.50%	31.50%
Claude Sonnet 4.5	15	0.0081	0.0237	0.00%	16.50%
Claude Sonnet 4.5	20	0.0078	0.0197	0.00%	16.50%
Claude Sonnet 4.5	30	0.0073	0.0150	0.00%	0.00%
Gemini 2.5 Pro	10	0.0167	0.0471	0.50%	48.50%
Gemini 2.5 Pro	15	0.0131	0.0350	0.00%	28.50%
Gemini 2.5 Pro	20	0.0110	0.0227	0.00%	10.50%
Gemini 2.5 Pro	30	0.0089	0.0151	0.00%	0.00%
GPT-4.1	10	0.0063	0.0263	0.00%	20.00%
GPT-4.1	15	0.0054	0.0150	0.00%	5.00%
GPT-4.1	20	0.0065	0.0149	0.00%	0.50%
GPT-4.1	30	0.0060	0.0114	0.00%	0.00%

21 Seed-Effect Empirical Test

The journal’s guidance for GenAI submissions requires authors to document specific random-seed values and, for commercial models, API versions and access dates; it stops short of establishing whether a seed parameter delivers determinism once documented.

Table 22: Seed effect vs. deployment effect. Cloud models tested: Claude Sonnet 4.5 (no seed parameter), Gemini 2.5 Pro and GPT-4.1 (seed=42). Local models all use seed=42.

Group	Stage	Mean EMR	Mean pairwise dis.
Cloud, seed=42	screening	0.953	0.017
Cloud, no seed	screening	0.974	0.010
Local, seed=42	screening	1.000	0.000
Cloud, seed=42	extraction	0.175	0.489
Cloud, no seed	extraction	0.050	0.818
Local, seed=42	extraction	1.000	0.000

Seed effect for extraction (cloud-seeded vs. cloud-unseeded): +0.125 EMR. Deployment effect (local-seeded vs. cloud-seeded): +0.825 EMR. The deployment effect is **6.6× larger** than the seed effect. We conclude that the seed parameter, as currently implemented in commercial APIs, does not deliver determinism; deployment paradigm dominates.

22 Pre-Registration Disclosure

This study is partly pre-registered, and the boundary is stated precisely. The original experiments (main text §4.1–§4.6) are exploratory: 10 repetitions, whole-output hash EMR, four semantic-tier definitions, and fixed-effect meta-analytic pooling were chosen during analysis design but without an OSF or AsPredicted submission. The dual-human labeling validation is the exception and *is* formally pre-registered as a frozen OSF registration dated 2026-05-12 (<https://doi.org/10.17605/OSF.IO/FGN3E>), two months before either label set was returned; see Section 4.2. The remaining post-revision analyses (§4.7 fixed-slot, §4.8 desconfound, §5.5 cost-benefit, silver-external) were designed and committed to the public Git repository *before* result computation; we do not present Git commits as equivalent to pre-registration.

Key commit timestamps (publicly auditable at <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>):

- **2026-02-11** (75292f2): project initialization, prompts committed
- **2026-02-13 to 2026-03-02**: original experiment raw outputs committed
- **2026-03-02** (1b90c1b): `analysis/reproducibility_results.json` committed (results frozen)
- **2026-03-11** (646c6a3): first complete manuscript draft
- **2026-03-20** (7935445): submission-hardening revision
- **2026-04-23 to 2026-04-28**: post-revision blindage analyses committed

We acknowledge the absence of formal pre-registration for the exploratory core as a limitation and recommend that future LLM-in-SR studies pre-register analyses on OSF before data collection.